

Yilin “Harry” Lu

Portfolio: <https://sites.google.com/umn.edu/yilinharry-lu/bio>

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EDUCATION

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- University of Minnesota, Twin Cities** Minneapolis, MN
 - Bachelor of Arts with Honors; GPA: 3.8/4.0* August 2023 - December 2026
 - Majors & Minor:** Triple Majors in Computer Science, Mathematics, Quantitative Economics (Minor: Statistics).
 - Honor Thesis (Computer Science):** *Visual Exploration and Explanation of Graph Neural Networks.*
 - Capstone Paper (Mathematics):** *Extending Partial Orders: From Binary Relations to Linear Orderings*
 - Capstone Paper (Quantitative Economics):** *Geometry and Topology on Scientific Innovation Networks.*
 - Honors:** Undergraduate Research Scholarship (\$ 1,800), Top 1 Paper at the Economic Writing Competition (\$ 750), Top 3 Project at the Travelers Predictive Modeling Competition, University Honor Program, Dean's List (2025, 2024, 2023).
 - Graduate-level Coursework:** Machine Learning Theory, Visualization for Interpretable AI, Cryptology, Predictive Analytics for Finance, Bayesian Statistics for Astrophysics, Machine Learning for Healthcare, Theory of Probability.
 - Cornell University** Ithaca, NY
 - Visting Student in Economics; GPA: 4.3/4.3* Winter 2025
 - San Diego State University** San Diego, CA
 - Concurrent Enrollment in Mathematics while in the High School* September 2022 - May 2023

ACADEMIC EXPERIENCE

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- University of Pennsylvania** Philadelphia, PA
 - Research Project Lead (Fully-funded by Penn): Human Computer Interaction, AI4Research* May 2026 - Present
 - Research Area(s):** Simulating Programmer's Behavior for DSL Researchers.
 - Northwestern University** Evanston, IL
 - Research Intern: Natural Language Processing, AI4Science, Science of Science* May 2026 - Present
 - Research Area(s):** Benchmarking Human-AI Complementary in Scientific Research.
 - University of Minnesota, Twin Cities** Minneapolis, MN
 - Research Project Lead: Explainable AI, Human Computer Interaction, Visualization* Jan 2024 - Present
 - Research Area(s):** Explainable AI for Non-expert Users [P3-P6, T1-T5], AI Agents and their Applications [P2, P8].
 - College of William & Mary** Williamsburg, VA
 - Research Intern: Natural Language Processing, AI4Science, AI4Research, AI Agents* Dec 2025 - Feb 2026
 - Research Area(s):** Autonomous AI for Scientific Research [P7], AI Agents and their Applications.
 - Minnesota Institute for Astrophysics** Minneapolis, MN
 - Research Intern: Human Computer Interaction, AI4Science, Astrophysics* September 2023 - May 2024
 - Research Area(s):** Crowdsourcing Systems for Scientific Discovery [P1], Image Processing for Transient Detections.

INDUSTRIAL EXPERIENCE

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- Microsoft Research Asia** Beijing, China
 - Incoming Research Intern (Societal AI, Human Computer Interaction)* Jan 2027 - May 2027
 - Beijing Dream Blossom Technology (formerly iQiYi Smart)** Beijing, China
 - Software Engineering Intern (Virtual Reality, UI Engineering)* July 2022 - August 2022
 - Responsibilities:** Developed a motion capture software in Unity 3D on Oculus VR platform; designed and animated 3D character models in Blender; prototyped virtual reality user interface in Unity 3D.

PUBLICATIONS

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- [P12] Simulating Programmer's Behaviors for Domain Specific Language Researcher.** : *Under Preparation to The ACM Conference on Human Factors in Computing Systems (CHI 2027).* **Y. Lu**, A. Head.
 - [P11] Human-AI Complementary Benchmark: Collaborative Decision Making in Scientific Research:** *Under Preparation to International Conference of Learning Representations (ICLR 2027).* Z. Guo, **Y. Lu**, J. R. Hullman.
 - [P10] GNN Explorer: Visual Exploration of Graph Neural Networks in Your Computational Notebooks:** *In Submission to The Association for the Advancement of Artificial Intelligence (AAAI 2026 Demo).* **Y. Lu**, Q. Wang.
 - [P9] Measuring Temporal Linguistic Emergence in a Diffusion Language Model:** *In Submission to The European Conference of Artificial Intelligence (IJCAI-ECAI 2026), Workshop on Explainable Artificial Intelligence.* **Y. Lu**.
 - [P8] Macro and Micro Uncertainty Signals for Prefix-Level Risk Prediction in Web Agents:** *In Submission to Empirical Methods of Natural Language Processing (EMNLP 2026).* S. Pan, Y. Shen, **Y. Lu**, Q. Wang.
 - [P7] SciDER: Scientific Data-centric End-to-end Researcher:** *In Submission to Empirical Methods of Natural Language Processing (EMNLP 2026).* K. Lin, O. Aijaz*, **Y. Lu***, Y. Luo, X. Guo, N. Presklov.
 - [P6] Play, Reflect, Understand: Developing a Web-Based Game for Learning Multi-Agent LLM Systems:** *In submission to ACM User Interface Software and Technology (UIST 2026).* **Y. Lu**, S. Du, Q. Wang.

- [P5] **GNN101: Visual Learning of Graph Neural Networks in Your Web Browser:** *IEEE Transactions on Visualization and Computer Graphics (TVCG)*. Y. Lu, C. Chen, Y. Chen, K. Huang, M. Zitnik, Q. Wang.
- [P4] **The Agentopia Times: Understanding and Mitigating Hallucinations in Multi-Agent LLM Systems via Data Journalism Gameplay:** *IEEE Visualization Conference (VIS 2025)*. Y. Lu, S. Du, Q. Wang.
- [P3] **What Can a Node Learn from Its Neighbors in Graph Neural Networks:** *IEEE Visualization Conference (VIS 2024), 7th VISxAI Workshop*. Y. Lu, C. Chen, M. Xu, Q. Wang.
- [P2] **What Helps Beyond Semantic Similarity? Utility-Driven Memory Selection for Robust Large Language Model Function Calling:** *In Submission to Empirical Methods of Natural Language Processing (EMNLP 2026)*. J. You, M. Gini, W. Zhu, S. Chen, Y. Lu, Y. Mao.
- [P1] **GRB231115A: TURBO Pre-Burst Optical Upper Limits:** *NASA General Coordinates Network (NASA GCN Circular)*. R. Strausbaugh et al., including Y. Lu (6th author).

JOURNAL REVIEW

- **Journal of Open Source Softwares:** Invited Reviewer (2026 - Present).
- **IEEE Trans. Visualization and Computer Graphics:** Invited Student Reviewer (2025 - Present).

CONFERENCE REVIEW

- **ACM Spatial User Interaction:** Conference Student Reviewer (2026).
- **ACM Automotive User Interface:** Conference Student Reviewer (2026).
- **ACM Creativity& Cognition:** Conference Student Reviewer (2026).

AUTHORED PACKAGES

- [L3] **kandor: An LLM-based World Building Engine:** A modularized LLM-based world building engine with a temporal knowledge graph memory system and an interactive visualization system for exploration in Python.
- [L2] **LargeGraphs.jl: Large Graph Sensemaking in Julia:** A webGL-based library (powered by Sigma.js) that makes large graphs explorable and explainable through interactive visualization in Julia, powered by WebGPU.
- [L1] **GNN Explorer: An Python Package for Visualizing Graph Neural Networks:** A package to visualizing different types of graph neural network for neuron-level interpretability and explainability.

PRESENTATIONS

- [T6] **Growth Pattern Analytics of Digital Infrastructures:** *UMN Undergraduate Research Workshop*. Y. Lu.
- [T5] **GNN 101:Visual Learning of Graph Neural Networks in Your Web Browser:** *IEEE VIS 2026*. Y. Lu.
- [T4] **The Agentopia Times: Understanding and Mitigating Hallucinations in Multi-Agent LLM Systems via Data Journalism Gameplay:** *IEEE VIS 2025*. Y. Lu.
- [T3] **The Agentopia Times: Understanding and Mitigating Hallucinations in Multi-Agent LLM Systems via Data Journalism Gameplay:** *UMN CS&E Graduate Student Orientation Poster Session (Invited)*. Y. Lu, S. Du.
- [T2] **GNN101: Visual Learning of Graph Neural Networks in Your Web Browser:** *UMN Visual Computing Seminar (Invited)*. Y. Lu.
- [T1] **What Can a Node Learn from Its Neighbors in Graph Neural Networks:** *IEEE VIS 2024*. Y. Lu.

PRESS

- **College of William & Mary pioneers AI scientist: SciDER enables machines to independently carry out the entire research process (in Mandarin):** *TechWalker*.
- **Excellence in Economic Writing: Celebrating Fall 2025 Award Winners:** *University of Minnesota*.

COMPUTER SCIENCE RESEARCH PROJECTS

- **DLEmerge: A Probing Framework for Temporal Linguistic Emergence of Diffusion Language Models:** A light-weight framework on probing how linguistic concepts emerge during inference processes in diffusion language models.
- **SciDER: Scientific Data-centric End-to-end Researcher [90 stars]:** A multi-agent system with a data processing agent, an experiment agent, and an ideation agent with self-evolve memory bank system, integrated in a Python package.
- **Agentopia: A Gamification Framework for Understanding Hallucination in LLMAS [5 stars]:** A customizable and grammar-driven game framework for understand and experiment hallucination in large language multi- agent systems.
- **ScholaVIS: A Toolkit for Intent Alignment Analysis of Human-AI Co-writing [5 stars]:** A scientific writing simulation tool integrated with different LLM agents, data analysis pipeline for event study, with a visual analytic software .
- **GNN101: An Interactive Visualization Tool for Graph Neural Networks [37 stars]:** An interactive tool for graph neural networks, developed in collaboration with Stanford and Harvard Medical School, deployed at Stanford CS224W and gained 1.6k+ users, designed a comparative study to evaluate the tool effectiveness against a baseline.

MATHEMATICS AND ECONOMICS RESEARCH PROJECTS

- **Extending Partial Orders: From Binary Relations to Linear Orderings:** A short note on ordering theory and set theory, specifically about how to extend binary relations to linear orderings.
- **Digital Drifting: Growth Pattern Analytics of Digital Infrastructures in the US: (Top 1 Project in the Econ Department)** A spatial-temporal convergence analysis of digital development in the US with a RShiny visualization.

RESEARCH MENTORSHIP

- **Shurui Du:** *B.Sc in Computer Science, UMN (Now: MSCS, Northwestern).* Co-authored Works: [P4, P6].
- **Yuxin Chen:** *B.Sc in Computer Science, UMN (Now: MSCS, UT Austin).* Co-authored Work: [P5].
- **Chongwei Chen:** *B.A with Honors in Mathematics, UMN.* Co-authored Works: [P3, P5].
- **Matthew Xu:** *B.Sc in Computer Science, UMN (Now: B.Sc. in Computing& Philosophy, UIUC).* Co-authored Work: [P3].

SKILLS SUMMARY

- **Languages:** Python, C/C++, TypeScript, JavaScript, Julia, R.
- **Visualization Technologies:** Unity 3D, D3.js, React.js, Phaser.js, IPython& IJulia, Flask, Streamlit, RShiny.
- **Machine Learning & Data Science:** Numpy, Pandas, Matplotlib, PyTorch, LangGraph, Scikit-learn, Transformers.
- **User Study:** Statistical Analysis, Experiment Design, Quantitative/Qualitative Analysis, Crowdsourcing.
- **Infrastructure& Deployment:** Serverless (Vercel, GitHub Pages), GPU Programming, AWS Deployment, Docker.
- **Agentic Automation:** Codex, Claude Agent SDK, Prompt Engineering, Harness Engineering (OpenClaw), autoresearch.

REFEREES

- **Qianwen Wang:** *Assistant Professor in Computer Science, University of Minnesota.* Email: qianwen@umn.edu
- **Andrew Head:** *Assistant Professor in Computer Science, University of Pennsylvania.* Email: head@seas.upenn.edu.
- **Jessica Hullman:** *Full Professor of Computer Science, Northwestern University.* Email: jhullman@northwestern.edu.